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A quantitative fuzzy-valued intersection matrix for obtaining fuzzy relationships between vague spatial objects

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ABSTRACT

Topological relation between geographical objects is one of the key components of the Geographical Information System (GIS). Considering the objects to be exact with sharp boundaries, four intersection, and nine intersection matrices initiated the study of calculating the topological relations. Due to the extensive existence of vague spatial phenomenons, the study of uncertainty object modeling has grown rapidly. Parallelly, the relationship calculating tools have been upgraded to handle vague concepts. The fuzzy 9-intersection matrix and Egg-Yolk methods are popular tools. Although the geographical objects are uncertain, surprisingly, most of these tools generate certain relations between them. The qualitative nature of these tools overlooks the essence of their uncertainty. We propose a quantitative fuzzy-valued 9-intersection matrix to overcome these drawbacks to obtain a fuzzy relationship between uncertain geographical objects. Then two new similarity computation techniques are introduced to calculate the membership grade between the proposed matrix and the known crisp matrices. These similarity computations allow two spatial objects to have partial membership against the eight established topological relations. The quantitative calculations indicate the strength of the relationship. The superiority of the proposed model is exhibited through comparative studies. Further, certain linguistic variables are linked to the evaluated membership grades to generate an immediate association with the known crisp relations. Examples are provided to demonstrate the applicability and efficacy of this association.

1. Introduction

Topological relation between geographical objects plays a pivotal role in the modeling and analysis of Geographical Information Systems (GIS). Initially, algebraic methods were introduced to determine the topological relations between the objects, assuming the objects as crisp sets. With the advancement of GIS research, it was established that spatial objects are not generally crisp. Instead, several types of uncertainty are occasionally associated with data during data mining, storing, and analyzing. Several studies, such as rough set, probability, and fuzzy set-based models, were introduced to manage uncertainty. Further studies [1,2] detected two major reasons for uncertainty: one is due to the positional inaccuracy, while the other is due to the inherent vagueness of the natural objects. In order to deal with the natural vagueness of spatial objects, researchers utilized the fuzzy set, whereas other techniques were used to deal with positional inaccuracy. The modeling of such vague objects is not straightforward [3] and several authors are working to generate and extract the vague spatial objects in the database systems. As the fuzzy set is a prominent tool for handling natural vagueness, several studies have considered vague spatial objects as fuzzy sets. A spatial database system with fuzzy objects is required to have a method to obtain the topological relation between those objects. Thus Bjørke [4], Tang and Kainz [5,6], Liu and Shi [7,8] and may more introduced intersection matrix based model to determine the topological relations. Several studies [4,9] specified various limitations in those models, such as the models are qualitative in nature, the significance of the obtained matrix is not known, only a single point of intersection was taken into consideration to generate the matrix elements and so on.

Considering the above facts, we introduce a novel fuzzy 9-intersection method, which can generate the quantitative relation between the objects by producing the fuzzy relation between the fuzzy objects. This paper primarily focuses on deriving the fuzzy relation between two simple fuzzy area objects. As the objects with spatial vagueness are considered as fuzzy sets, we denote such objects as fuzzy objects or vague spatial objects in this article, Whereas the term ‘uncertainty’ is used in a broader sense, which represents both inaccuracy and vagueness.

The study overcomes most of the issues that were raised during the previous studies. Rather than using a single point of intersection, our method considers the fuzzy aggregating operator, where all points of intersection are used to construct the elements of the intersection matrix. The fuzzy topological parts of the objects, namely the fuzzy interior, boundary, and exterior, are used to construct nine mutual intersections,

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which serve as the entries of a 3 × 3 matrix. We termed that matrix a fuzzy-valued 9-intersection matrix (FV9IM). Due to the consideration of fuzzy aggregating operator, the entries of the matrix belong to [0; 1]. Next, two similarity computations formulas are proposed for the fuzzy relationship between the objects, which measure the membership of the obtained FV9IM corresponding to the basic relationships. Relation between those similarity computation and their separate usability are discussed. The obtained fuzzy memberships are then further connected with linguistic variables in order to incorporate with the query-based languages. Finally, possible applications of the proposed theories are provided.

The remaining sections of the paper are structured as follows. Section 2 describes the preliminary concepts and previous studies. Section 3 proposes a model capable of describing the quantitative relationship between two uncertain objects and determining the relationship as fuzzy. Section 4 concludes the paper and proposes some future work.

2. Related work

This section reviews and analyzes the research related to relationship-obtaining models in spatial database management systems and fuzzy topology. We focus on the prominent studies which have laid the foundation for spatial relationship analysis in GIS.

Topological relation between the objects is a crucial concept in GIS, used for query, analysis, and to check data consistency. Intersection matrix models such as 4-intersections [10] and 9-intersections [11] are basic and efficient methods to compute the relation between two objects in GIS.

2.1. Topological relations between crisp objects

Initially, the 4-intersection method and 9-intersection methods [10–12] were introduced to obtain the topological relations between the spatial data types by assuming the spatial objects as the crisp sets. Those models employed point set topology to obtain topological properties, namely interior, boundary and exterior and constructed intersection matrices by calculating the quality (empty or non-empty) of the mutual intersections among those topological properties. The 9-intersection matrix between two crisp objects A and B was defined as follows:

$$\begin{pmatrix} \{\mathrm{Int}A \cap \mathrm{Int}B\} & \{\mathrm{Int}A \cap \mathrm{Bd}B\} & \{\mathrm{Int}A \cap B^{-}\} \\ \{\mathrm{Bd}A \cap \mathrm{Int}B\} & \{\mathrm{Bd}A \cap \mathrm{Bd}B\} & \{\mathrm{Bd}A \cap B^{-}\} \\ \{A^{-} \cap \mathrm{Int}B\} & \{A^{-} \cap \mathrm{Bd}B\} & \{A^{-} \cap B^{-}\} \end{pmatrix} \quad (1)$$

where IntA, BdA, A^{-} are respectively the interior, boundary, and exterior of a set A in a crisp topological space.

Mathematically, using this method is possible to get 2^9 different types of relation. Still, due to several conditions, the total number of realizable relations between two simple area objects are restricted to eight only, while the relation between simple area and simple line objects was found to be nineteen, and the total number of the relation between two simple line objects was thirty-three only. Table 1 describes all the eight relations and their corresponding 9-intersection matrices for two simple area objects.

Later, different techniques and generalizations of 9-intersection matrix such as Region Connection Calculus (RCC) models [13], the Voronoi-based 9-intersection matrix [14] were constructed to determine the topological relations between the crisp spatial objects. Till date, depending on the object space, several generalizations of 9-intersections matrices have been proposed, for example, 9+-intersection matrix [15], binary topological relations on the digital sphere [16,17], topological relations between directed line segments in the cyclic space [18], direction relation matrices [19], relation between fuzzy spatiotemporal objects based on XML (Extensible Markup Language) [20–22], topological relationships in R^3 for 3d Cadastre [23], generalized 9-intersection matrix for regions with holes [24], intuitionistic fuzzy 9-intersection matrix [25,26], topological relationship model for geographical flows [27].

2.2. Uncertain object modeling

The rigorous research revealed uncertainty as one of the major issues in the field of GIS. Despite extensive research, a definitive solution to address uncertainty in GIS remains elusive [3]. As a result, researchers pursue various approaches to manage uncertainty in GIS-based problems. In literature, a variety of approaches have been used to manage the issue of uncertainty. Probability theory has been employed to handle the uncertainty related to position and measure [28–32]. Rough set-based models [33–36] have been developed using the rough set theory. The fuzzy set has been used to manage and analyze vague spatial data [9,37], fuzzy decision making [38–40] and so on. Among these studies fuzzy set theory has emerged as an important branch in this field.

2.3. Fuzzy set for modeling uncertain objects in GIS

Due to its widespread applications in uncertain spatial analysis, the fuzzy set became a popular framework in GIS and related disciplines. Generalizing the concept of crisp set, Zadeh [41] introduced the concept of fuzzy set, where the belongingness of the elements is described using the membership function.

2.3.1. Fuzzy set

Definition 2.1. Let X ‘ be an ordinary set and I = [0; 1]. Let $_A$: X ™ I be a function for A Ï X, where $_A$ (or simply A(x)) is said to be the *membership function* of A for all x ∈ X. The set A along with the membership function is said to be the I *fuzzy set* on X.

For two fuzzy set A and B, few standard fuzzy set operations are as follows;

- Union $(A \vee B)(x) = \max[A(x); B(x)]$.
- Intersection $(A \wedge B)(x) = \min[A(x); B(x)]$.
- Complement $A^c(x) = 1 - A(x)$.

2.3.2. Fuzzy aggregating operator

The fuzzy aggregating operator is used to combine multiple fuzzy membership degrees into a single aggregated membership degree. The concept was initially introduced by Fung and Fu [42].

Definition 2.2. The aggregation operator f, for n fuzzy sets was defined as f : $[0;1]^n$ ™ [0; 1], where n ∈ N and n > 1.

Several studies [43–45] were conducted in order to create several types of application-driven fuzzy aggregating operators. Bloch [46] studied such operators to combine information for the purpose of data fusion. The study categorized the aggregation operators into three different types; An aggregation is conjunctive if $f(x_1; x_2; \ldots; x_n)$ f $\min(x_1; x_2; \ldots; x_n)$, disjunctive if $f(x_1; x_2; \ldots; x_n)$ g $\max(x_1; x_2; \ldots; x_n)$ and compromise if $\min(x_1; x_2; \ldots; x_n)$ f $f(x_1; x_2; \ldots; x_n)$ f $\max(x_1; x_2; \ldots; x_n)$, where $(x_1; x_2; \ldots; x_n) \in [0;1]^n$ and $f(x_1; x_2; \ldots; x_n) \in [0; 1]$. Further, the study classified the aggregation operators into four classes, e.g., (i) T-norms, (ii) T-conorms, (iii) Symmetric sum and (iv) Mean.

An overall review regarding the subject was performed by Mardani et al. [47] and Hossain et al. [48] discussed the performance of several fuzzy aggregating operators in various applications.

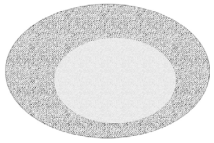
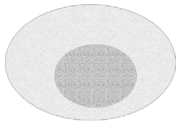
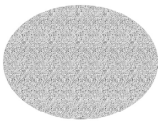
2.3.3. Fuzzy topology

Chang introduced the fuzzy topology [49] in 1968.

Definition 2.3. For a set X ‘ and I be the unit interval and Ï I^X be a collection of fuzzy sets, is said to be a fuzzy topology (FT) for X, if

(1) $0_X; 1_X \in$,
(2) A; B ∈ Ù A ∧ B ∈ ,

Table 1
9-intersection matrices for area–area relations.

Disjoint	Contains	Inside	Equal
			
$I_0 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$	$I_1 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$	$I_2 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$	$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
$I_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$	$I_5 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$	$I_6 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$	$I_7 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$

(3) $(A_j)_{j\in J} \in \tau \Rightarrow \bigvee_{j\in J} A_j \in \tau$, where J is an index set.

The members of τ are called open fuzzy sets, and complement of elements of τ are said to be closed fuzzy sets.

For a fuzzy set A in a FT space, the interior of A, denoted by $IntA$, is defined as the supremum of all fuzzy open set contained in A, the closure of A, denoted by ClA is defined as the infimum of the all fuzzy closed set that contains A, and the exterior (Ext) of A, denoted as $\bar{A}$, is defined as $\bar{A} = (ClA)^c$.

Each of the definitions in the above paragraph is due to [49].

2.3.4. Fuzzy boundary

The fuzzy boundary of a fuzzy set in a FT space is an important aspect that requires special attention in this study. Contrary to the crisp case, the boundary of a fuzzy set can have a non-empty intersection with the interior and the exterior of the set. Actually, it almost mimics the attributes of geographical objects found in nature. Those objects hardly have any distinguishable boundary.

In the present literature, we have different definitions [50–54] of the fuzzy boundary, as listed below. Among all these definitions, the one introduced by Pu and Liu [51] is mostly used in the literature.

Definition 2.4. The boundary of A denoted by BdA, defined as $BdA = ClA \wedge ClA^c$.

2.4. Application of fuzzy set in GIS

The fuzzy set theory finds its wide range of applications in GIS related problems. So far it has been applied for fuzzy spatial object modeling [1,37], fuzzy modeling of natural objects using field observation [55,56], qualitative spatiotemporal reasoning using fuzzy logic [57–59], remote sensing of GIS data [38,60], sensitivity analysis [39,40,61], fuzzy risk assessment modeling [62–64], multi-criteria decision analysis [65,66] and so on. Despite the fact that the majority of these studies utilized the concept of fuzzy sets, they have primarily focused on a modeling perspective. But, these studies have not extensively addressed the handling or representation of fuzzy spatial objects in a spatial database system.

2.5. Fuzzy spatial objects

The initial models in the study field were developed by Altman [67]. The study by Dilo [2] and Schneider [1] provided a deeper analysis of the uncertain spatial object. The reason for uncertainty was categorized into two; One type was recognized as geometric uncertainty, which is triggered due to the positional inaccuracy of the crisp objects. The other type of uncertainty is due to the vague spatial nature of the natural objects, which was recognized as thematic uncertainty or spatial vagueness. The object types are identified as spatial objects with vague boundaries. There are several definitions of simple fuzzy objects are available in the literature; Schneider [1], Tang and Kainz [5], Liu and Shi [68], Carniel and Schnider [69], and many more has defined simple fuzzy objects. In this study we follow the definition given by [1], where the simple fuzzy spatial objects, namely; simple fuzzy point, line and area objects were defined as follows:

Definition 2.5. A fuzzy point $P(a;b)$ is defined by $P_{(a;b)}(x;y) = \begin{cases} \text{if } (x;y) = (a;b) \\ 0 \quad \text{otherwise} \end{cases}$, where $0 < \text{f} \leq 1$ and $P(a;b) \in R^2$

Definition 2.6. L is said to be a simple fuzzy line, if the membership function of L is defined by $L : f \to [0;1]$ and $f : [0;1] \to R^2$ and the functions L and f satisfies the following conditions.

(i) Both L and f are continuous,
(ii) $\forall\, x;y \in (0;1); x \neq y \Rightarrow f(x) \neq f(y)$,
(iii) $\forall\, x \in \{0;1\}\ \forall y \in (0;1); f(x) \neq f(y)$,
(iv) $f(0) < f(1) \vee (f(0) = f(1) \wedge \forall x \in (0;1); f(0) < f(x))$.

In order to define a fuzzy region Spatially regular fuzzy subspace (SRFS) was introduced by Schneider [9]. Special union, intersection, difference and negation operators were introduced to retain the closure property of the SRFS. Then a fuzzy region was regarded as the collection of fuzzy faces

Definition 2.7. R is said to be a fuzzy region if

(i) $R = \dot{\bigcup}_{i=1}^{n} C_i; n \in N$,
(ii) $\forall\, 1 \leq i \leq n : C_i$s are connected components,
(iii) $R = \dot{\bigcup}_{i=1}^{n} C_i$,
(iv) $\forall\, 1 \leq i \leq n$: where each of the C_i is a piecewise continuous fuzzy faces.

Rather than using the term 'fuzzy' or 'uncertain' object the study by Dilo [2] refer the fuzzy objects or the objects with vague boundary as 'vague objects'. Dilo [70], defined simple vague objects, such as simple vague point, line, and area objects. The definitions of simple vague point and vague line are similar to the definition of [1]. The study define the vague region as follows

Definition 2.8. A fuzzy set A is said to be a vague region if A is bounded, $A = Cl(IntA)$, ans $IntA$ is connected.